Name

Tonya Hendricks

Date:

Fall 2023 Semester

Instructor:

Samuel Chukwuemeka

Project:

Number Theory Application: Check Digits for International Standard Book Numbers (ISBNs)

Book Title:

Introduction to Contemporary Education: New Horizons

Copyright Year, Edition, and ISBNs:

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2nd Edition

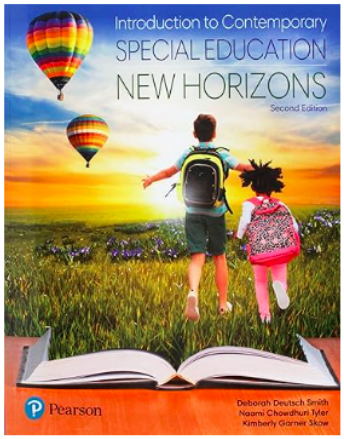
ISBN-10: 0134895088

ISBN-13: 978-0134895086

Objectives:

(1.) Verify the check digit for ISBN-10
(2.) Verify the check digit for ISBN-13
(3.) Convert ISBN-10 to ISBN-13
(4.) Convert ISBN-13 to ISBN-10

Calculations:



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Introduction to Contemporary Special Education: New Horizons 2nd Edition

by Deborah Smith (Author), Naomi Tyler (Author), Kimberly Skow (Author)

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0134895088	978-0134895086	2nd	Pearson	March 1, 2018

Objective 1: Verify the check digit for ISBN-10

Given: 0134895088

Approach: Multiplication-Modulo-Subtraction Approach

$0 \times 10 = 0$

$1 \times 9 = 9$

$3 \times 8 = 24$

$4 \times 7 = 28$

$8 \times 6 = 48$

$9 \times 5 = 45$

$5 \times 4 = 20$

$0 \times 3 = 0$

$8 \times 2 = 16$

Sum of the Products

$0 + 9 + 24 + 28 + 48 + 45 + 20 + 0 + 16 = 190$

Sum divided by Modulo

$190 \div 11 = 17\ Remainder\ 3$

$Modulo\ -\ Remainder$

$11 - 3 = 8$

This is the correct check digit.

<u>Objective 2: Verify the check digit for ISBN-13</u>

Given: 978-0134895086

Approach: Multiplication-Modulo-Subtraction Approach

$9 \times 1 = 9$

$7 \times 3 = 21$

$8 \times 1 = 8$

$0 \times 3 = 0$

$1 \times 1 = 1$

$3 \times 3 = 9$

$4 \times 1 = 4$

$8 \times 3 = 24$

$9 \times 1 = 9$

$5 \times 3 = 15$

$0 \times 1 = 0$

$8 \times 3 = 24$

Sum of the Products

$9 + 21 + 8 + 0 + 1 + 9 + 4 + 24 + 9 + 15 + 0 + 24 = 124$

Sum Divided by 10

$124 \div 10 = 12\ Remainder\ 4$

$Modulo - Remainder$

$10 - 4 = 6$

This is the correct check digit.

<u>Objective 3: Convert ISBN-10 to ISBN-13</u>

Given: 0134895088

Step 1: Remove check digit: 013489508

Step 2: Append 978: 978013489508

Step 3: Calculated check digit in 2nd objective. The check digit is 6.

ISBN-13: 9780134895086

<u>Objective 4: Covert ISBN-13 to ISBN-10</u>

Given: 9780134895086

Step 1: Remove the check digit: 978013489508

Step 2: Remove the 978: 013489508

Step 3: Calculated check digit in 1st objective. The check digit is 8.

ISBN-10: 0134895088

References (MLA9):

Chukwuemeka, Samuel. "Number Theory Application: Check Digits for ISBN." *Mathematics Education Courses*, 2023, mathematicseducation.appspot.com/.

Smith, Deborah Deutsch, et al. *Introduction to Contemporary Special Education: New Horizons*. 2nd ed., Pearson, 2018.